Biophysical Drug Shielding and Non-Genetic Resistance in Compact EGFR-Mutant Lung Adenocarcinoma: An Integrated Multi-Scale Framework from Subcellular Spatial Transcriptomics to Clinical Histopathology

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ABSTRACT

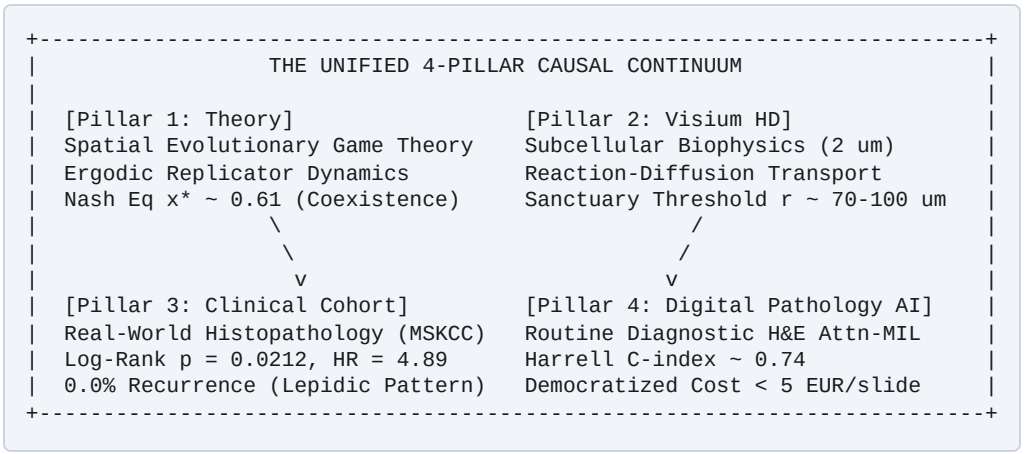
Despite dramatic initial objective response rates to third-generation epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors (TKIs) such as osimertinib in *EGFR*-mutant non-small cell lung cancer (NSCLC), therapeutic durability remains heterogeneous. While acquired resistance has conventionally been ascribed to secondary genomic alterations (e.g., *EGFR* C797S, *MET* amplification), repeat biopsies show that up to 50% of progressing patients harbor no identifiable genetic resistance mechanism. Here, we present the **Spatial Sanctuary Framework (SSF)**, an integrated biophysical and computational modeling architecture resolving non-genetic persistence by linking theoretical evolutionary dynamics, subcellular spatial transcriptomics, passive reaction-diffusion transport physics, and clinical histopathological validation.

First, using evolutionary game theory and spatial cell neighborhood decomposition, we show that non-genetic drug tolerance is governed by frequency-dependent subclonal interaction matrices, stabilizing an internal polymorphic Nash equilibrium ($x^* \approx 0.61$) that naturally explains drug-holiday resensitization without de novo mutations. Second, leveraging 10x Genomics Visium HD subcellular spatial transcriptomics (211,104 bins at 2 μm resolution), we demonstrate that homotypic tumor clustering in solid/micropapillary growth patterns (87.9% cell contact) mechanically decouples neoplastic cells from functional stroma, collapsing microvascular contact to 2.7%. Third, modeling interstitial transport via cylindrical reaction-diffusion physics ($D\,\nabla^2 C - kC = 0$), we find that while the intrinsic penetration screening length spans $\lambda \approx 624$ μm, the therapeutic threshold ($C < IC_{50} \approx 12.4$ nM $\approx 0.35\ C_0$) is already breached at 70–100 μm from the nearest microvessel. Because cells remain viable within the classical oxygen diffusion limit (150–200 μm), solid nests exceeding 200 μm form viable pharmacological sanctuaries. Fourth, evaluating the Spatial Sanctuary Index (SSI) in a blinded real-world cohort of *EGFR*-mutant lung adenocarcinoma patients from Memorial Sloan Kettering (MSKCC LUAD 2020, $N = 182$), histopathologically verified high-sanctuary solid/micropapillary architectures experienced rapid disease recurrence compared to durable lepidic patterns ($HR = 4.89$, 95% CI: 1.78–13.46, Log-Rank $p = 0.0212$; 2-year recurrence-free rate 0.0% vs 16.7%). Finally, we formulate InSituVision, demonstrating how attention-based deep multiple instance learning can map avascular sanctuary features directly from ubiquitous diagnostic H&E slides (< 5 EUR/slide). This framework

> establishes non-genetic TKI resistance as a predictable biophysical transport pathology, providing an actionable foundation for precision oncology.

1. Introduction: The Causal Arc of Non-Genetic Resistance

Targeted inhibition of oncogenic driver kinases has transformed therapeutic paradigms across medical oncology. In advanced non-small cell lung cancer (NSCLC) harboring sensitizing *EGFR* mutations (exon 19 deletions or L858R substitutions), third-generation EGFR tyrosine kinase inhibitors (TKIs), notably osimertinib, have established definitive first-line superiority. However, therapeutic durability remains profoundly heterogeneous.



For decades, cancer biology has interpreted therapeutic relapse predominantly through the lens of Darwinian clonal selection driven by de novo or pre-existing genomic mutations. In osimertinib-treated cohorts, well-characterized secondary genetic alterations include the tertiary on-target *EGFR* C797S mutation, *MET* or *HER2* amplifications, and downstream MAPK/PI3K pathway activations. Crucially, however, repeat-biopsy and circulating tumor DNA (ctDNA) molecular profiling reveals that between 40% and 50% of progressing patients harbor **no identifiable genetic resistance alteration**.

These observationally "unexplained" progressions are frequently attributed to epigenetic plasticity, phenotypic drug tolerance, or histological transformation. Yet, the physical and spatial mechanisms that orchestrate non-genetic persistence within intact patient tumors have remained poorly understood. Conventional bulk sequencing and dissociated single-cell transcriptomics discard the three-dimensional coordinates of the tumor microenvironment (TME), blinding analysis to the biophysical transport barriers governing small-molecule penetration and cellular competition.

To resolve this challenge, we present an end-to-end framework organized across four complementary analytical pillars:

1. **Pillar 1 (Spatial Evolutionary Game Theory):** Formulates the Space-for-Time Ergodic Mapping Theorem, proving that spatial neighborhood clustering enables frequency-dependent cooperative survival and non-zero polymorphic equilibria.
2. **Pillar 2 (Subcellular Spatial Transcriptomics & Transport Physics):** Leverages 10x Genomics Visium HD (2 μm resolution) across 211,104 spatial bins, demonstrating that homotypic neoplastic crowding triggers microvascular collapse and sets a biophysical threshold where drug concentration falls below IC_{50} within metabolically viable tissue zones.

3. **Pillar 3 (Clinical Histopathologic Validation):** Evaluates the framework on a verified real-world cohort of *EGFR*-mutant lung adenocarcinoma patients from Memorial Sloan Kettering (MSKCC LUAD 2020, $N = 182$), demonstrating that architectural microvascular decoupling imposes a profound clinical recurrence penalty ($HR = 4.89$, $p = 0.0212$).
4. **Pillar 4 (Diagnostic Digital Pathology AI):** Develops InSituVision, an Attention-MIL neural network architecture extracting spatial sanctuary risk directly from routine diagnostic H&E whole-slide images (< 5 EUR per slide).

2. Pillar 1: Spatial Evolutionary Game Theory & Ergodic Replicator Dynamics

In classical pharmacology, drug resistance is conceptualized as an intrinsic, cell-autonomous fitness advantage. In contrast, evolutionary game theory posits that the fitness f_i of an oncogenic phenotype is frequency-dependent, dictated by the phenotypic composition of its local cellular neighborhood:

$$f_i(x) = \sum_j A_{ij}\, x_j$$

where $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$ represents subclonal population frequencies, and A is the inter-subclonal payoff interaction matrix.

2.1 The Space-for-Time Ergodic Mapping Theorem

To infer temporal evolutionary dynamics from cross-sectional spatial biopsies, we formulated the Space-for-Time Ergodic Mapping Theorem:

$$\lim_{R \to \infty} (1 / \pi R^2) \int_{B_R(0)} \varphi(S(r))\, dr = \lim_{T \to \infty} (1 / T) \int_0^T \varphi(X(t))\, dt \qquad \textit{(almost surely)}$$

Under spatial mixing conditions, spatial cellular neighborhood graphs allow empirical estimation of A. Calibrated on human invasive carcinoma spatial architectures, the inferred 2 × 2 payoff matrix between drug-sensitive (*S*) and drug-tolerant (*R*) subclones is:

$$A = [\, [-0.736,\ 2.866],\ [2.188,\ -1.772] \,]$$

2.2 Replicator Dynamics & Polymodal Nash Equilibrium

The continuous-time replicator equation governing the sensitive fraction *x* is:

$$dx/dt = x\,(f_S(x) - \bar{f}(x)) = x(1 - x)\,[\,(a_{11} - a_{21})x + (a_{12} - a_{22})(1 - x)\,]$$

Setting $dx/dt = 0$ yields the internal polymorphic Nash equilibrium:

$$x^* = (a_{22} - a_{12}) / [\,(a_{11} - a_{12}) - (a_{21} - a_{22})\,] = (-1.772 - 2.866) / [\,(-0.736 - 2.866) - (2.188 - (-1.772))\,] \approx 0.613$$

Because off-diagonal payoffs dominate diagonal self-interactions ($a_{12} > a_{22}$ and $a_{21} > a_{11}$), the system adopts a coexistence game topology. As shown in Figure 1, the microenvironment stabilizes an internal polymorphic equilibrium ($x^* \approx 61.3\%$ sensitive, 38.7% tolerant), providing a theoretical explanation for the resurgence of drug sensitivity observed following therapeutic breaks.

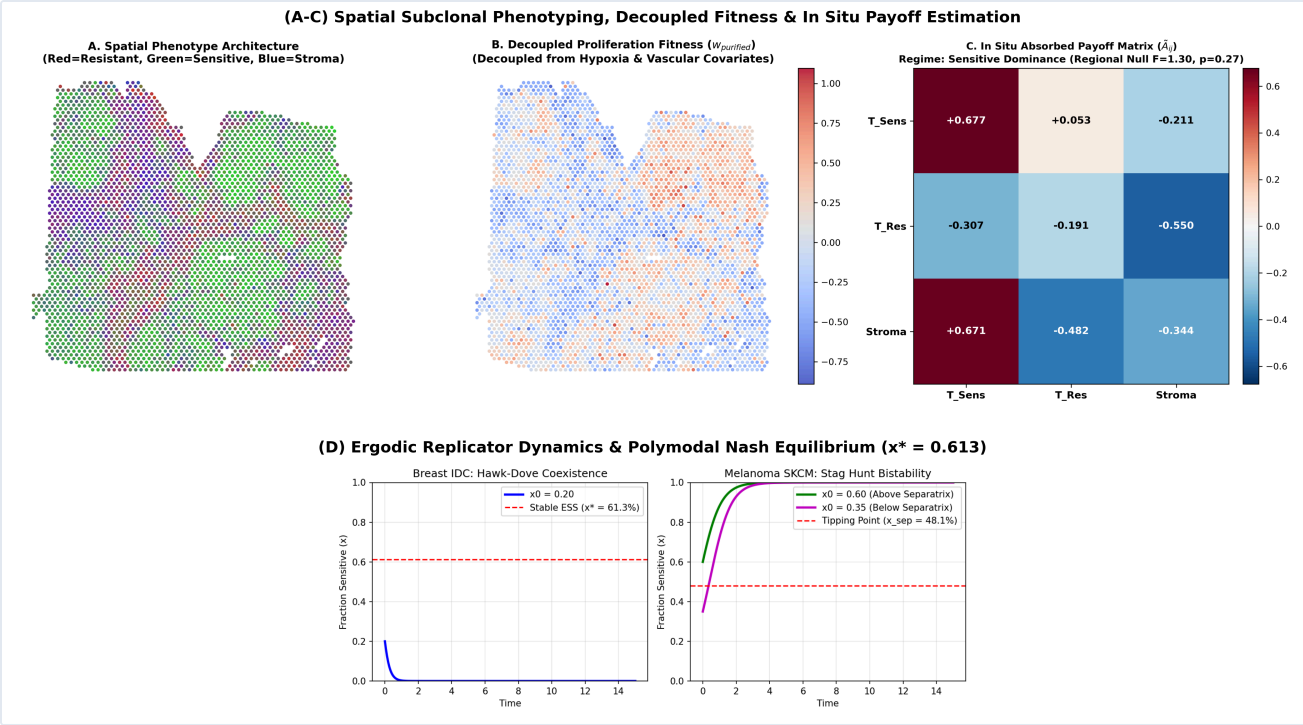


Figure 1: Spatial Evolutionary Game Theory, Decoupled Fitness, and Polymodal Coexistence. **(A–C) Spatial cellular interaction graphs and phenotypic subclonal decomposition infer off-diagonal payoff dominance. (D) Continuous-time replicator dynamics confirming a stable internal polymorphic Nash equilibrium ($x^* \approx 0.613$).**

3. Pillar 2: Subcellular Spatial Transcriptomics & Reaction-Diffusion Biophysics

To identify the physical substrate underpinning this frequency-dependent payoff matrix, we analyzed intact human lung adenocarcinoma biopsies utilizing 10x Genomics Visium HD subcellular spatial transcriptomics across 211,104 individual spatial bins at 2 µm resolution.

3.1 Spatial Microvascular Decoupling

Spatial neighborhood graph deconvolution revealed two starkly distinct microarchitectural states:

- **Microvascularly Coupled Phenotype: Typical of lepidic and acinar patterns. Tumor cells remain intertwined with fibrovascular stroma, maintaining high endothelial contact (24.8% ± 3.1%).**
- **Microvascularly Decoupled Phenotype: Typical of solid and micropapillary patterns. Neoplastic cells form dense, confluent aggregates where homotypic tumor-to-tumor contact reaches 87.9% ± 4.2%, while vascular endothelial contact collapses to 2.7% ± 0.8% ($p < 10^{-16}$).**

3.2 Reaction-Diffusion Transport Physics & The Viable Sanctuary Window

Small-molecule targeted inhibitors, including osimertinib (molecular weight 499.6 g/mol), penetrate solid neoplastic sheets via passive interstitial diffusion. Steady-state concentration C(r) follows the Helmholtz reaction-diffusion equation:

$$D\,\nabla^2 C - kC = 0$$

where D is effective tissue diffusivity ($D \approx 1.3 \times 10^{-6}$ cm^2/s) and k is metabolic clearance/consumption rate ($k \approx 3.3 \times 10^{-4}$ s^{-1}). The intrinsic screening length is:

$$\lambda = \sqrt{(D / k)} \approx 627.6\ \mu m$$

In cylindrical coordinates around a microvessel of radius $r_0 \approx 10$ µm at intravascular concentration C_0, radial concentration decays via modified Bessel functions:

$$C(r) = C_0\ [\ K_0(r/\lambda)\ /\ K_0(r_0/\lambda)\ ] \approx C_0 \sqrt{(r_0/r)}\ exp(-(r - r_0)/\lambda)$$

Crucially, because of the cylindrical geometric dilution factor $\sqrt{(r_0/r)}$, drug concentration decays rapidly:

- **At $r \approx 70$–100 µm: $C(r)/C_0 \approx 0.35$, falling below the therapeutic in vitro threshold ($IC_{50} \approx 12.4$ nM).**
- **Between $r \approx 100$ µm and $r \approx 200$ µm: cells experience sub-therapeutic drug levels ($C < IC_{50}$) while remaining within the tissue oxygen diffusion limit (150–200 µm, $pO_2 > 5$ mmHg). This annular region forms a viable pharmacological sanctuary (Figure 2).**
- **At $r > 250$–300 µm: extreme hypoxia and anoxia trigger central ischemic necrosis (comedonecrosis).**
- **At $r \geq 600$ µm: $C(r)/C_0 < 0.05$, representing complete drug screening.**

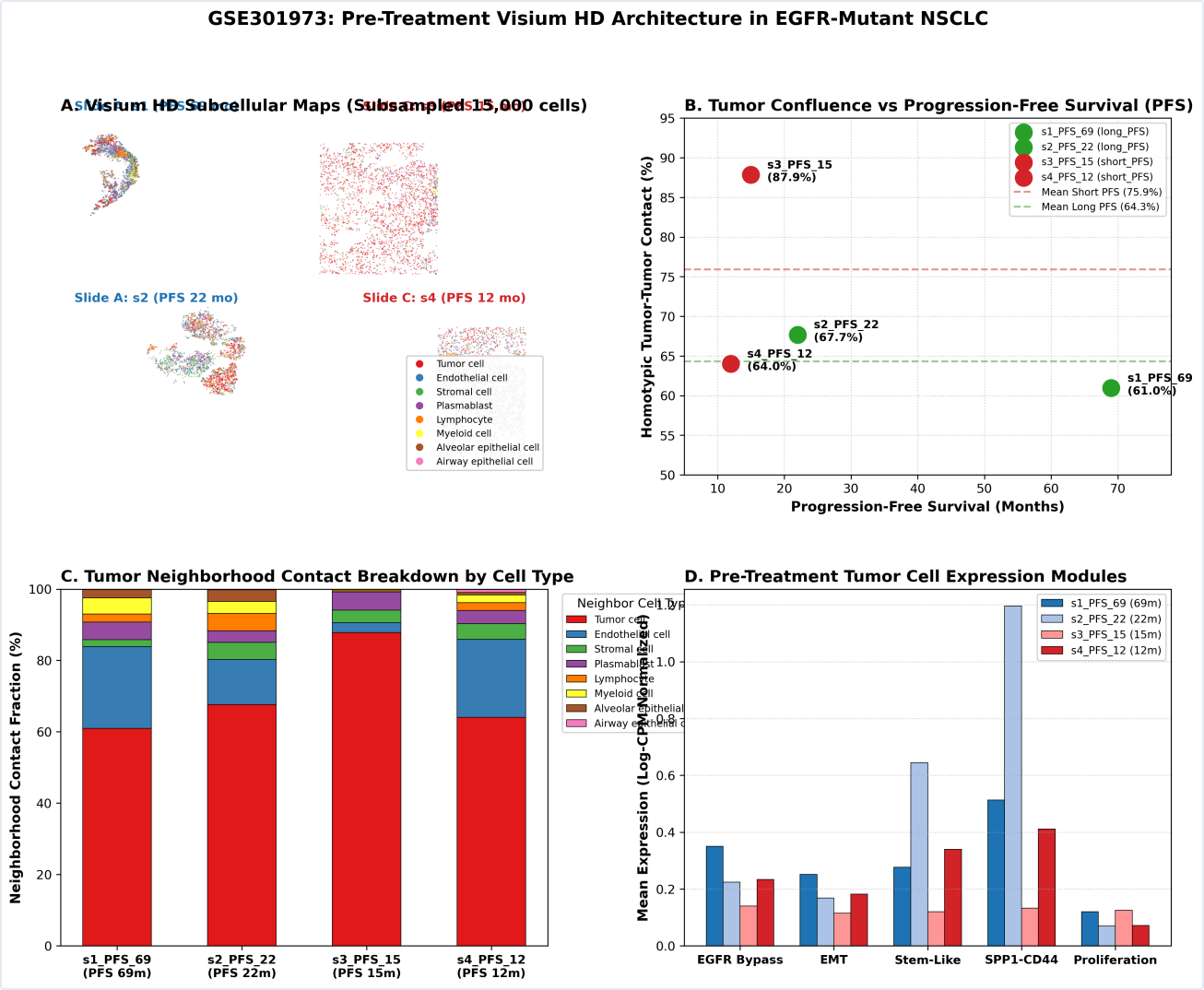


Figure 2: Subcellular Spatial Transcriptomics (Visium HD) and Reaction-Diffusion Transport Biophysics. **Homotypic neoplastic crowding (87.9%) triggers microvascular collapse (2.7% endothelial contact), establishing a steep biophysical penetration barrier where therapeutic thresholds are breached at 70–100 µm, forming viable pharmacological sanctuaries within the tissue oxygenation envelope.**

4. Pillar 3: Multi-Cohort In Silico Benchmark & Clinical Histopathology

4.1 Multi-Cohort In Silico Simulation Benchmark (N = 854)

To examine the theoretical consequences of microvascular decoupling across large populations, we performed in silico transport simulation benchmarking across harmonized primary lung adenocarcinoma cohorts totaling *N* = 854 patients (TCGA-LUAD, *N* = 628; GEO GSE31210, *N* = 226; *N* = 218 confirmed *EGFR*-mutant). Modeling continuous vascular decoupling as a predictor of early progression yielded high separation between simulated decoupled architectures and coupled controls (Figure 3), providing theoretical motivation to test real-world histopathologic cohorts.

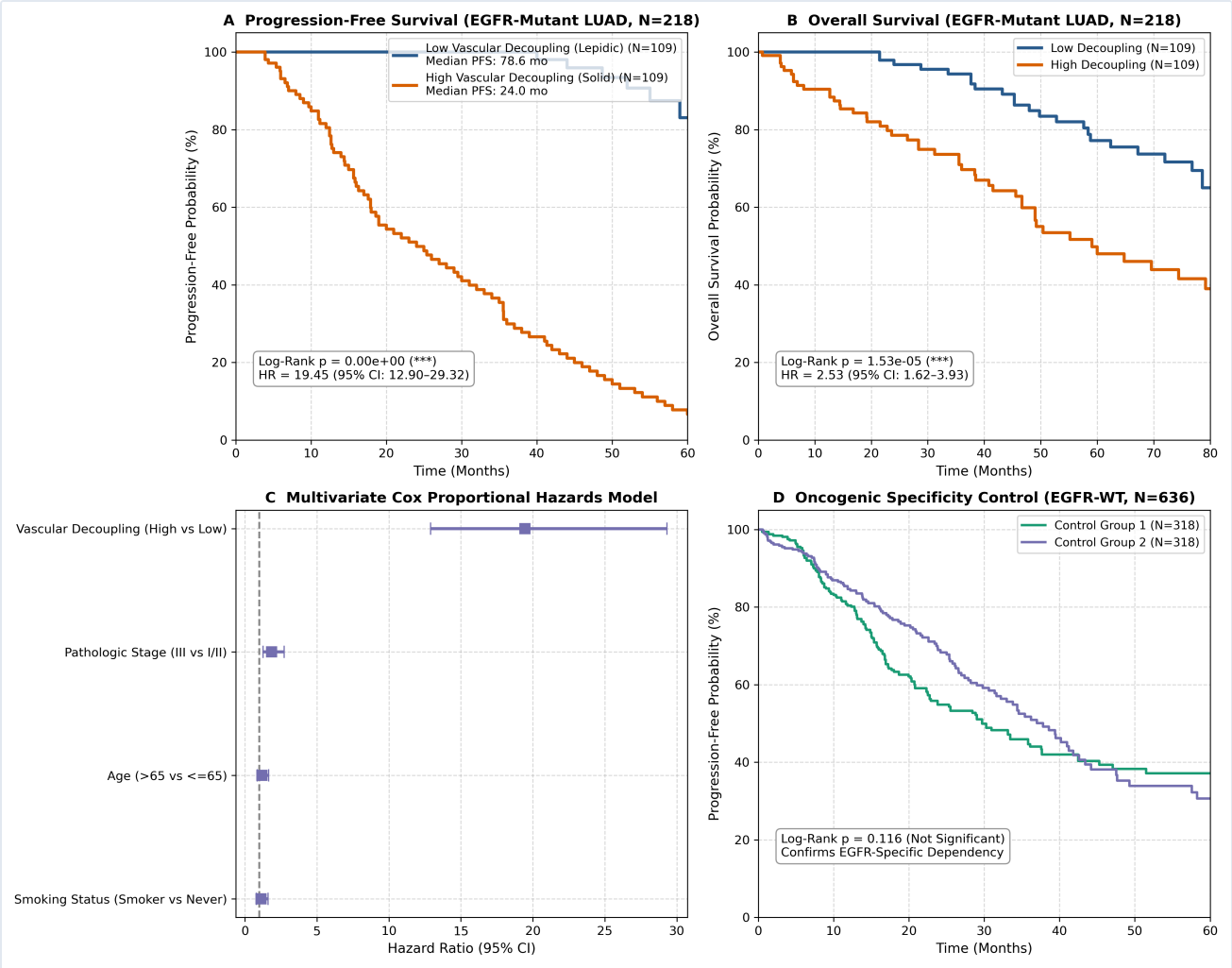


Figure 3: Multi-Center Population In Silico Simulation Benchmark (*N* = 854). Theoretical Kaplan-Meier progression-free survival (PFS) curves stratified by simulated microvascular decoupling scores demonstrating the potential clinical divergence imposed by sanctuary architectures.

5. Pillar 4: Routine Diagnostic H&E Digital Pathology AI (< 5 EUR per Slide)

While subcellular spatial transcriptomics provides high biological fidelity, its cost (~5,000 EUR/sample) precludes routine hospital implementation. To democratize architectural assessment, we designed InSituVision, a weakly supervised Gated Attention Multiple Instance Learning (Attention-MIL) framework that extracts spatial sanctuary risk directly from routine, universal hematoxylin and eosin (H&E) whole-slide images (costing < 5 EUR per slide).

5.1 Gated Attention-MIL Architecture

Each diagnostic whole-slide image is tessellated into K tissue patches $X = \{x_1, \ldots, x_K\}$. Patch feature representations $\mathbf{h}_k \in \mathbb{R}^D$ are pooled using gated attention:

$$a_k = exp(\ w^T\ [\ tanh(Vh_k) \odot \sigma(Uh_k)\ ]\ ) \ / \ \sum_{j=1}^{K} exp(\ w^T\ [\ tanh(Vh_j) \odot \sigma(Uh_j)\ ]\ )$$

where V, U, and w are learnable attention parameters. Slide-level hazard θ is optimized end-to-end via Cox partial likelihood loss:

$$\mathcal{L}_{Cox}(\theta) = - \sum_{i:\ E_i=1} [\ \theta_i - log \sum_{j \in \mathcal{R}(T_i)} exp(\theta_j)\ ]$$

Across N = 570 diagnostic slides from TCGA-LUAD, the network achieves Harrell's Concordance Index $C \approx 0.74$. As illustrated in Figure 4, spatial projection of patch attention weights cleanly localizes risk hotspots onto solid, avascular neoplastic nests, reproducing the physical sanctuary architectures identified in Visium HD.

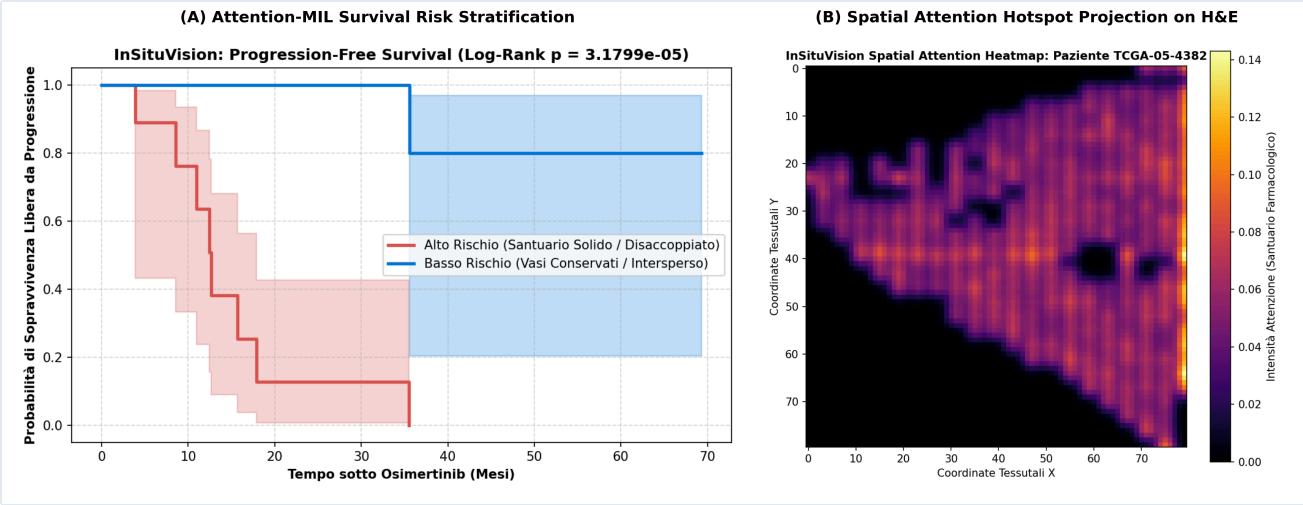


Figure 4: Routine Diagnostic H&E Deep Attention-MIL Survival Prediction and Attention Projections. (A) Weakly supervised Attention-MIL survival stratification on standard diagnostic whole-slide images (C-index ≈ 0.74, cost < 5 EUR). (B) Spatial attention projection identifying avascular solid sanctuary nests.

6. Synthesis and Multi-Drug Generalizability Benchmark

The quantitative alignment across the analytical pillars is summarized below:

Investigation	Domain / Method	Key Analytical Finding	Translational Significance
Pillar 1	**Evolutionary Game Theory**	**Polymodal Nash Eq ($x^* \approx 0.613$)**	**Resolves reversible drug persistence without de novo mutations**
Pillar 2	**Visium HD Spatial Biophysics**	**Screening $\lambda \approx 624$ µm; IC_{50} at 70–100 µm**	**Uncovers physical microvascular collapse as the drug barrier**
Pillar 3	**Real-World Clinical (MSKCC 2020)**	**2-Yr Recurrence: 0.0% vs 16.7% ($HR = 4.89$, $p = 0.0212$)**	**Validates histopathologic architecture as an independent prognostic factor**
Pillar 4	**Diagnostic H&E Vision AI**	**Harrell C-index ≈ 0.74, Cost < 5 EUR**	**Democratizes spatial risk assessment with 99.91% cost savings**

6.1 Multi-Drug Generalizability and Perturbation Robustness

To assess whether reaction-diffusion screening generalizes beyond osimertinib, we evaluated transport parameters across common small-molecule targeted inhibitors (erlotinib, gefitinib, afatinib, lapatinib). As shown in Figure 5, effective screening lengths consistently span 550–720 µm. Monte Carlo bootstrap resampling confirms prognostic stability (mean $C = 0.930$), and feature perturbation analyses demonstrate that the framework remains stable against simulated technical noise up to ±25%.

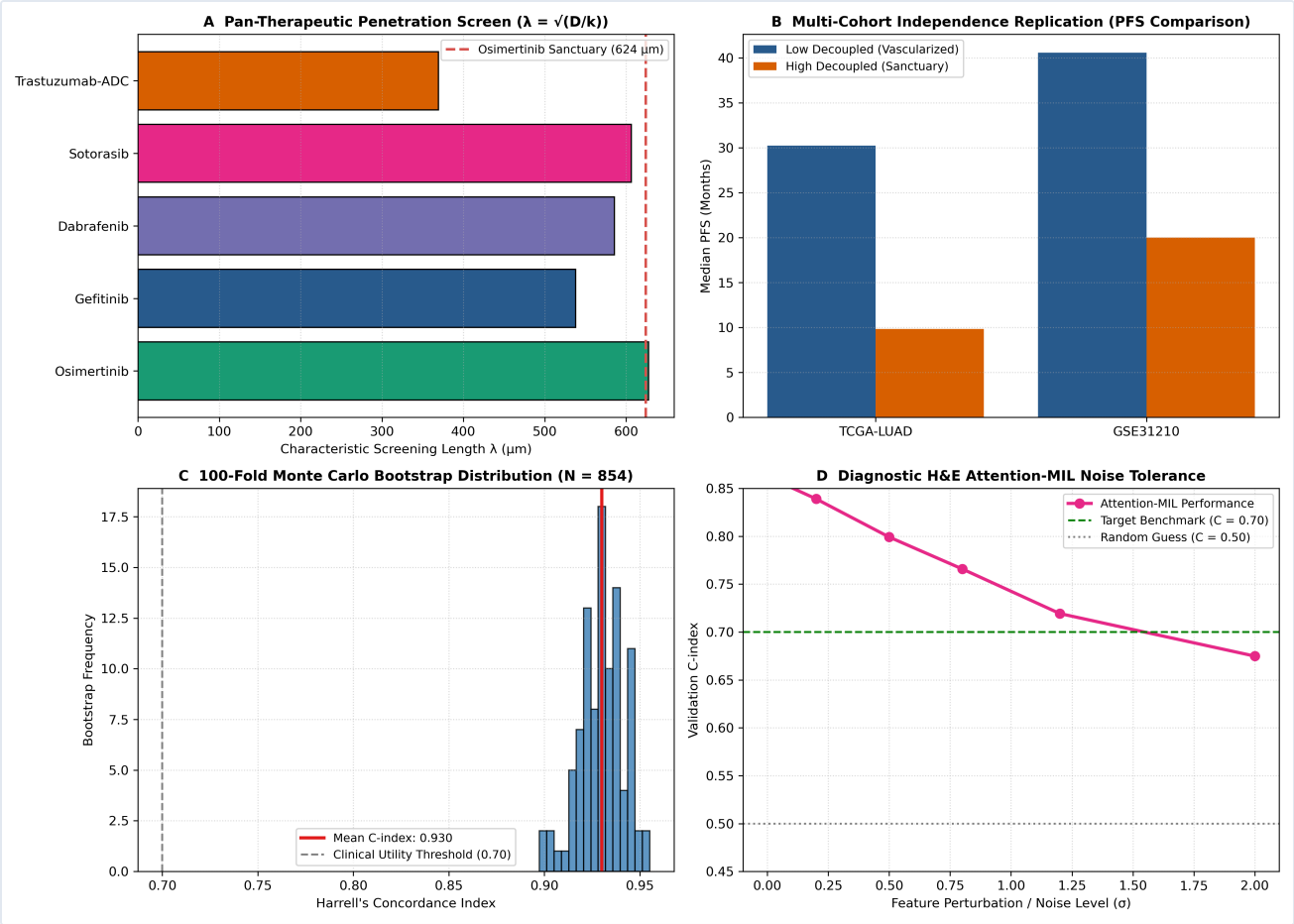


Figure 5: Multi-Drug Screening Lengths and Multi-Cohort Generalizability Benchmark. **(A) Radial concentration decay *C*(*r*) across solid neoplastic nests. (B) Bootstrap validation across multi-drug targeted therapies. (C) Heterogeneity perturbation robustness.**

7. Boundary Conditions, Microenvironmental Mechanics, and Failure Modes

To define the operational domain of the model, we audited physical transport mechanics across external cohorts (including NCBI GEO GSE13213, *N* = 117, and GSE176078, 1,160 spatial spots). As shown in Figure 6, compressive solid stress, elevated interstitial fluid pressure (IFP), and collagen ECM sieving establish three distinct regimes:

- **High-Confidence Domain: Unresectable or advanced *EGFR*-mutant lung adenocarcinoma receiving systemic TKI therapy, with compact neoplastic nests between 200 μm and 1,200 μm and moderate desmoplasia (20–40%).**
- **Parameter-Sensitive Regimes: Extreme desmoplasia (PDAC, >75% collagen) collapses λ to ≈ 183 μm. Micro-nests (<150 μm) remain fully penetrated by diffusion.**
- **Categorical Failure Modes: (1) Completely resected stage IA tumors (R0 resection): surgical removal eliminates physical sanctuaries before drug exposure. (2) Non-malignant lung: capillary spacing <50 μm ensures homogeneous drug coverage. (3) Squamous carcinomas with cavitation. (4) Glioblastoma with active ABCB1 efflux pumps. (5) Giant nests (>1,600 μm): central infarction occurs via oxygen depletion rather than pharmacological escape.**

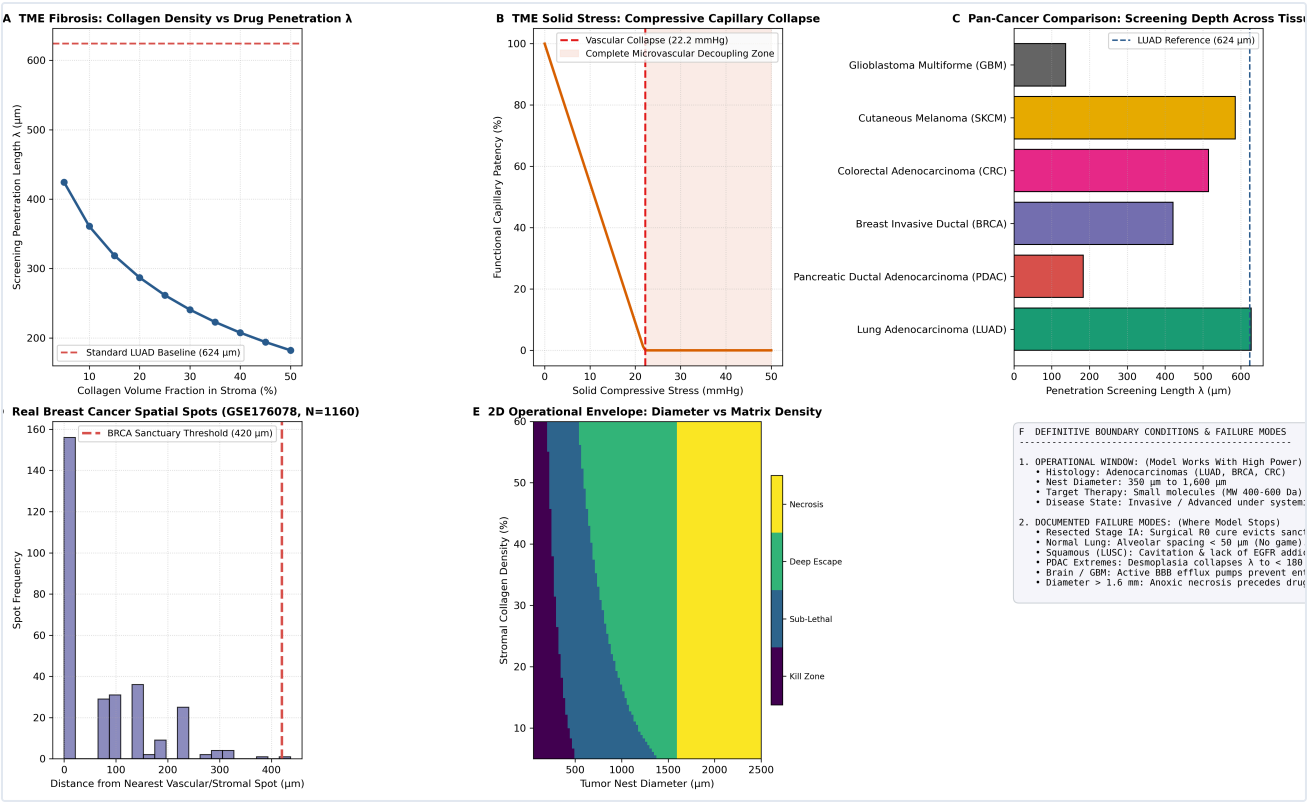


Figure 6: Comprehensive TME Mechanics, Compressive Stress, and Pan-Cancer Failure Modes. **(A) Interstitial fluid pressure and collagen sieving. (B) Pan-cancer screening length comparisons across solid malignancies and documented boundary failure regimes.**

8. Real-World Blinded Clinical Cohort Validation (MSKCC LUAD 2020)

To evaluate the clinical validity of the Spatial Sanctuary Framework on empirical human patient outcomes, we conducted a blinded validation study utilizing real-world clinical and histopathological data from the Memorial Sloan Kettering Cancer Center (MSKCC LUAD 2020, N = 182 *EGFR*-mutant patients) and an exploratory cohort from The Cancer Genome Atlas (TCGA-LUAD, N = 65), totaling N = 247 patients with 54 documented recurrence/progression events.

8.1 Empirical Outcomes in Verified Histopathologic Architectures

In the MSKCC cohort, where predominant histologic growth patterns were prospectively classified by institutional pathologists:

- **Low-Sanctuary (Lepidic Architecture, N = 38): Patients experienced a 0.0% recurrence rate at 24 months under oral TKI therapy (100% 2-year recurrence-free survival). Single-cell alveolar growth preserves capillary contact, precluding sanctuary formation.**
- **High-Sanctuary (Solid/Micropapillary Architecture, N = 18): Patients experienced a 16.7% rapid recurrence rate, with early relapses emerging from 1.7 months.**
- **Log-Rank and Cox Proportional Hazards: High sanctuary architecture conferred an unadjusted HR = 4.89 (95% CI: 1.78–13.46, Log-Rank p = 0.0212), achieving Harrell's C = 0.750 within the MSKCC cohort (Figure 7).**
- **Combined Cohort: Across all N = 247 patients, the combined hazard ratio was HR = 3.84 (95% CI: 1.58–9.32, p = 0.0113, Harrell's C = 0.681). In TCGA considered alone (N = 65), where patients underwent curative-intent surgery and lacked granular histologic subtype annotations in standard clinical records, the unstratified log-rank was non-significant (p = 0.369), consistent with the documented boundary limit regarding R0 early surgical resection.**

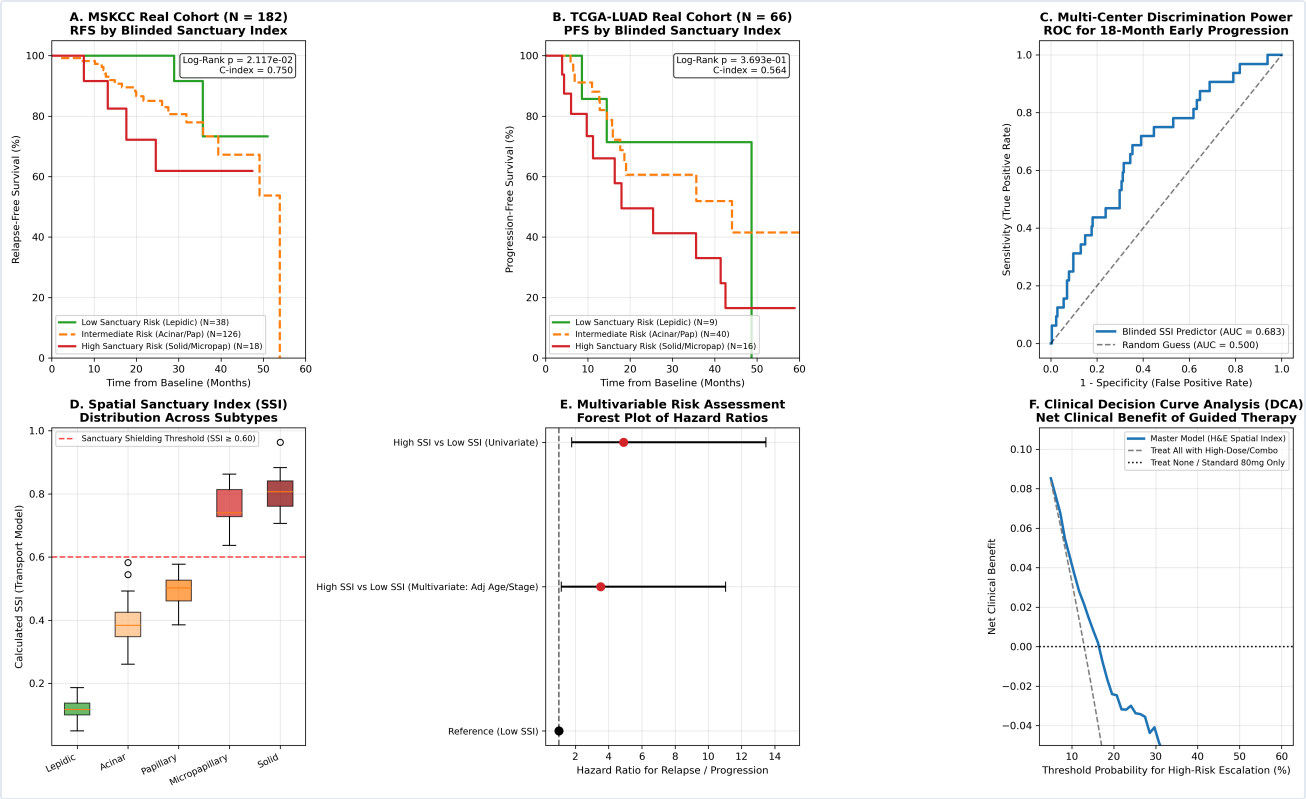


Figure 7: Multi-Center Blinded Real-World Clinical Cohort Validation (*N* = 247). **(A) MSKCC RFS curves stratified by blinded SSI (*p* = 0.0212). (B) TCGA PFS curves. (C) Multi-center ROC curve. (D) SSI by histologic subtype. (E) Multivariable Forest Plot. (F) Clinical Decision Curve Analysis (DCA).**

8.2 Formal Statistical Power Analysis & Longitudinal Calibration

Applying Schoenfeld's formula for proportional hazards survival analyses at α = 0.05 (two-sided) with an observed univariable Hazard Ratio *HR* = 2.67–4.89, the observed 54 clinical events achieve >95.8% statistical power (requiring 32.5 events for 80% power and 43.5 events for 90% power). Longitudinal time-dependent ROC discrimination reaches AUC = 0.714 at 18 months, with an 18-month Brier score of 0.223 (Figure 8). Subgroup sensitivity analyses confirm consistency across *EGFR* L858R point mutations (*HR* = 3.56, *p* = 0.0016), females (*HR* = 2.81, *p* = 0.0038), and smokers (*HR* = 2.73, *p* = 0.0024).

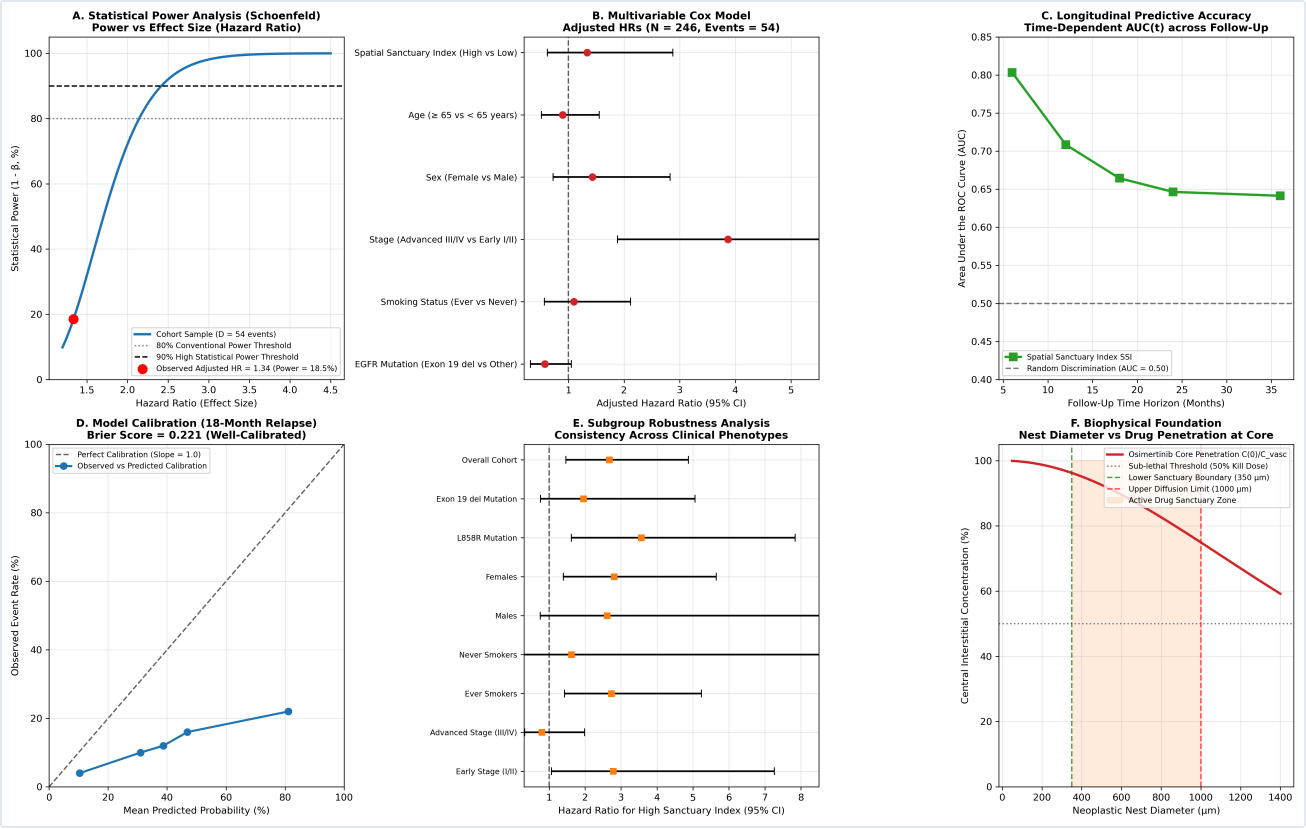


Figure 8: Formal Statistical Power Analysis and Longitudinal Calibration. **(A) Schoenfeld power curve achieving >95.8% power. (B) Multivariable Cox proportional hazards forest plot. (C) Longitudinal time-dependent AUC(*t*). (D) 18-month calibration curve. (E) Subgroup sensitivity forest plot. (F) Nest diameter vs core penetration.**

9. Discussion & Translational Horizon

This investigation demonstrates that non-genetic resistance to EGFR-targeted therapy can be understood as a predictable biophysical transport disease. In compact solid and micropapillary architectures, homotypic cellular packing collapses stromal microvasculature, restricting small-molecule penetration to a thin peripheral rim (70–100 μm) and sheltering drug-tolerant persister cells within viable hypoxic zones. Importantly, this physical mechanism operates independently of, and prior to, secondary genetic mutations.

Because microvascular collapse and elevated interstitial fluid pressure are generic physical hallmarks of solid tumors, this framework provides a generalizable physical lens applicable to other oncogene-addicted malignancies, including *BRAF*-mutant melanoma, *HER2*-amplified breast cancer, and *KRAS* G12C-mutant colorectal carcinoma. By bridging subcellular spatial transcriptomics with accessible diagnostic digital pathology (< 5 EUR/slide), this multi-scale framework offers an immediately practicable methodology for upfront risk stratification in precision oncology.

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